Data Description

1. customers.csv (5 Columns)

Column	Data Type	Description
customer_id	String	Unique identifier for each customer (UUID). Serves as the primary key in the Customer table.
became_member_on	Integer YYYYMMDD	Date on which the customer created their account (e.g., 20170212 means February 12, 2017). Should be converted to Date in Power BI.
gender	String	Customer's self-reported gender: M, F, or O (Other). May contain nulls if undisclosed.
age	Integer	Customer's age in years. Some records use 118 to indicate "age missing/unknown"—treat these as null or flag for imputation.
income	Float	Estimated annual income in USD (e.g., 112000.0). Null if unknown or not provided.

Notes on customers.csv:

- Convert became_member_on from integer (YYYYMMDD) to a proper Date type.
- Replace out-of-range or placeholder ages (e.g., 118) with null or a separate "unknown" category.
- For missing gender or income, you may choose to impute (median income by age bracket) or create a "Not Disclosed"/"Unknown" category.

2. offers.csv (6 Columns)

Column	Data Type	Description
offer_id	String	Unique identifier for each marketing offer (UUID). Primary key for the Offer dimension.
offer_type	String	Type of offer (one of: bogo = buy-one/get-one, discount, or informational).
difficulty	Integer	Minimum purchase amount required to unlock the reward (e.g., 20 means the customer must spend $20 to receive a discount).
reward	Integer	Dollar value of the reward (e.g., $5 off). Always zero for informational offers.
duration	Integer	Number of days the customer has to complete the offer once they receive it (e.g., 7 means 7 days from the “offer received” event).
channels	String	List of marketing channels through which the customer can receive the offer (e.g., ['email','mobile','social']). Stored as a JSON array.

Notes on offers.csv:

- channels is a JSON-formatted array of strings. In Power Query, split or unpivot this into a lookup table if you want to analyze channel effectiveness separately.
- difficulty = 0 and reward = 0 for informational offers (no purchase requirement, simply educational).
- Join on offer_id when combining with the Events table.

3. events.csv (4 Columns)

Column	Data Type	Description
customer_id	String	Foreign key referencing customers.customer_id. Indicates which customer triggered this event.
event	String	Type of event. Possible values:
		• offer received – When an offer was delivered to the customer.
		• offer viewed – When the customer opened or viewed the offer.
		• offer completed – When the customer fulfilled the offer criteria (e.g., spent the required amount).
		• transaction – When the customer made a non-offer transaction (purchase) in the system.
value	String	A JSON string whose contents depend on the event type:
		• For offer received, offer viewed, offer completed: {'offer id': '<offer_id>'}
		• For transaction: {'amount': <numeric value>} (e.g., {'amount': 12.34}).
		Use Power Query's JSON parsing functions to extract offer_id or amount into separate columns.
time	Integer	Number of hours (0–719) since the start of the 30-day observation window for that customer.
		E.g., 0 means the very first hour after the customer became a member; 100 means 100 hours later.

Notes on events.csv:

- time is relative to each customer's "start of observation" (which is often the moment they became a member).
- To analyze event chronology, convert time to an actual timestamp by adding time (in hours) to became_member_on (converted to DateTime at midnight).

- Split value into two new columns in Power Query:
    1. offer_id_extracted – if the event is related to an offer (received/viewed/completed).
    2. transaction_amount – if the event is a transaction.

---

4. data_dictionary.csv (3 Columns)

Column	Data Type	Description
Table	String	Name of the source table (offers, customers, events) that the following Field belongs to.
Field	String	Exact column name in that table.
Description	String	Textual explanation of what that field represents (pulled from dataset documentation).

Usage of data_dictionary.csv:

- This file contains human-readable definitions for every field in the three core tables.
- You can use it to build tooltips or summary documentation inside Power BI.
- Example entries:
    - For offers.offer_id: “Unique Offer ID (primary key)”
    - For customers.became_member_on: “Date when the customer created their account (YYYYMMDD)”
    - For events.event: “Description of the event (purchase, offer received, etc.)”